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Paleomagnetism of Jurassic sediments as evidence for oroclinal bending of the Inner West Carpathians

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ABSTRACT

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The Jurassic carbonates from the Križna Nappe contain a pre-tectonic characteristic component of remanence. The declinations for individual nappe fragments differ, and form a fan-like pattern. They are close to the tectonic vergencies measured in the field, which suggests that the arcuate structure of the Inner Carpathians results from an oroclinal bending of an originally straight belt.

Introduction

The Carpathian Fold Belt forms the most important and complex arcuate structure of the East European part of the Alpine chain. The problem of the origin of the observed curvatures is still unresolved. Paleomagnetic data may help to determine whether the curvatures in a given region are original or the result of an oroclinal bending, i.e. a tectonic deformation of an originally straight belt. Lowrie and Hirt (1986) showed how the relationship between the declination deviation and fold axis deviation helps to resolve such problems in arcuate mountain belts. A similar analysis was conducted by Bazhenov and Burtman (1990) for the Cretaceous rocks of the whole Carpathian Belt from the Southern Alps to the Eastern Pontides. Their results suggest an oroclinal bending of originally parallel straight belts. The present paper reports the results of a paleomagnetic study of the Križna Nappe Jurassic sediments exposed in the Inner West Carpathians.

The Carpathian Fold Belt is divided into the Outer and Inner Carpathians by the Pieniny Klippen Belt (Fig. 1). The Jurassic sediments occur in the Inner Carpathians as allochthoneous nappes, which probably originated at the northern margin of the Tethys in the vicinity of the Čertovica Line in the Low Tatras (Bely and Fusan, 1967). During the Mediterranean phase of the Alpine orogeny these sediments were detached from their basement, pushed northward and folded. As a result of tectonic processes they were divided into three nappes lying one upon the other. The lowermost and best preserved Križna Nappe lies directly on the Hightatric substratum, and its structure is probably closely related to the structure of the substratum rocks. The northward pushing and folding of the nappe resulted in its fragmentation into several separate nappe fragments. The Križna Nappe sediments comprise rock formations from the Permian Verrucano (rarely exposed), through the Triassic and the Jurassic, up to the Lower Cretaceous. The Jurassic sediments, which are

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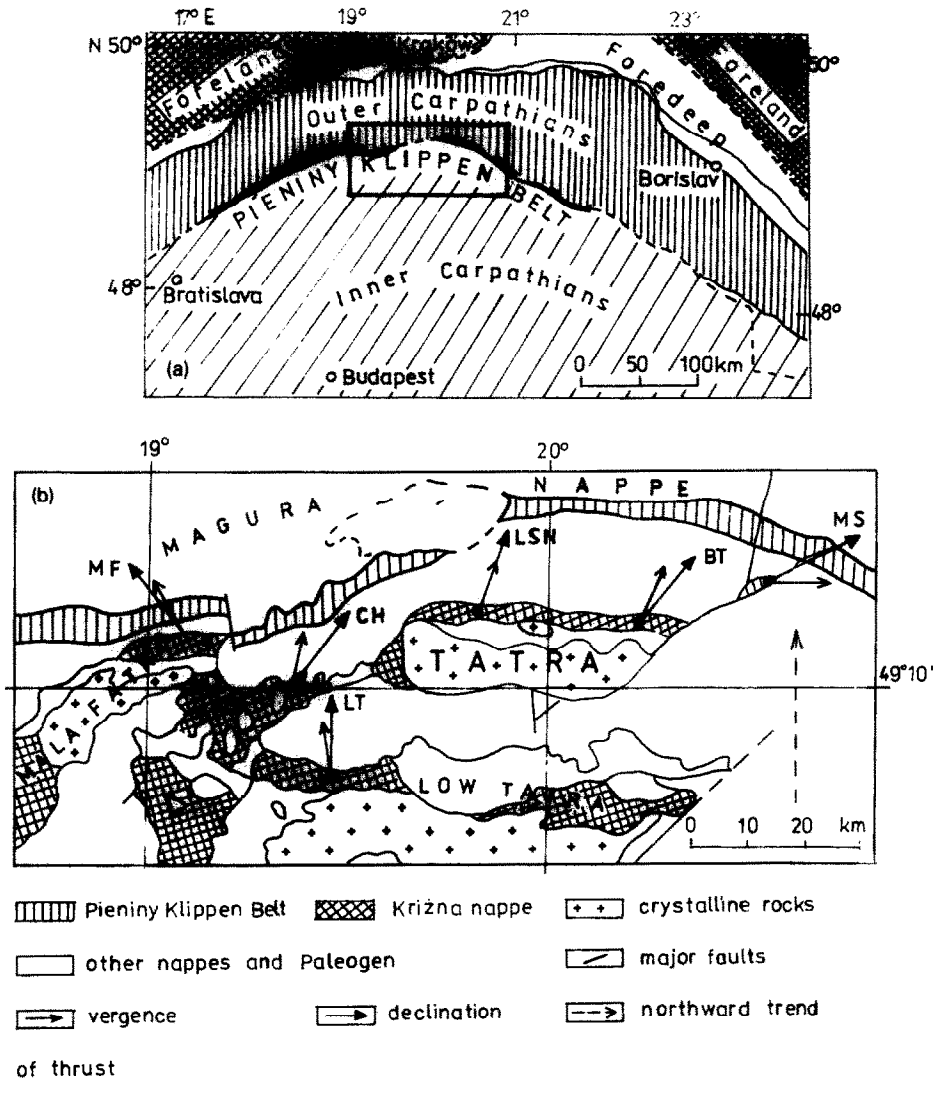


Fig. 1. Simplified geological map of the area. (a) The position of the investigated region (rectangle) in the West Carpathians. (b) A simplified map of the investigated part of the Inner West Carpathians. The relevant nappes are denoted as follows: *MF* = Mala Fatra Mts.; *CH* = Choč Hills; *LT* = Low Tatra Mts.; *LSN* = Lower Subtatric Nappe; *BT* = Belanski Tatra Mts.; *MS* = Magura Spišska. Circles denote the sampled exposures.

well developed carbonates and occasionally silicites, have been studied in detail (Lefeld, 1974) and their biostratigraphic age, based on aptychi, is well determined (Gąsiorowski, 1962). They provide good material for the study of the structure of the region.

Geology and sampling

Six of the nappe fragments in the Križna unit (Fig. 1b) have been studied paleomagnetically. The results for the Mala Fatra Mts. nappe have been given by Kądziałko-Hofmokl et al. (1990), and those for the Lower Subtatric Nappe by Kądziałko-Hofmokl and Kruczyk (1987). Each nappe fragment has a different bedding orientation (see Table 1) and their vergencies (Fig. 1b) form a fan-like pattern around the northward trend of overthrusting. This paper presents the results from four nappe fragments of the Križna unit; the Choč Hills (CH), the Lower Tatra (LT), the Belanski Tatra (BT) and Magura Spišska (MS)

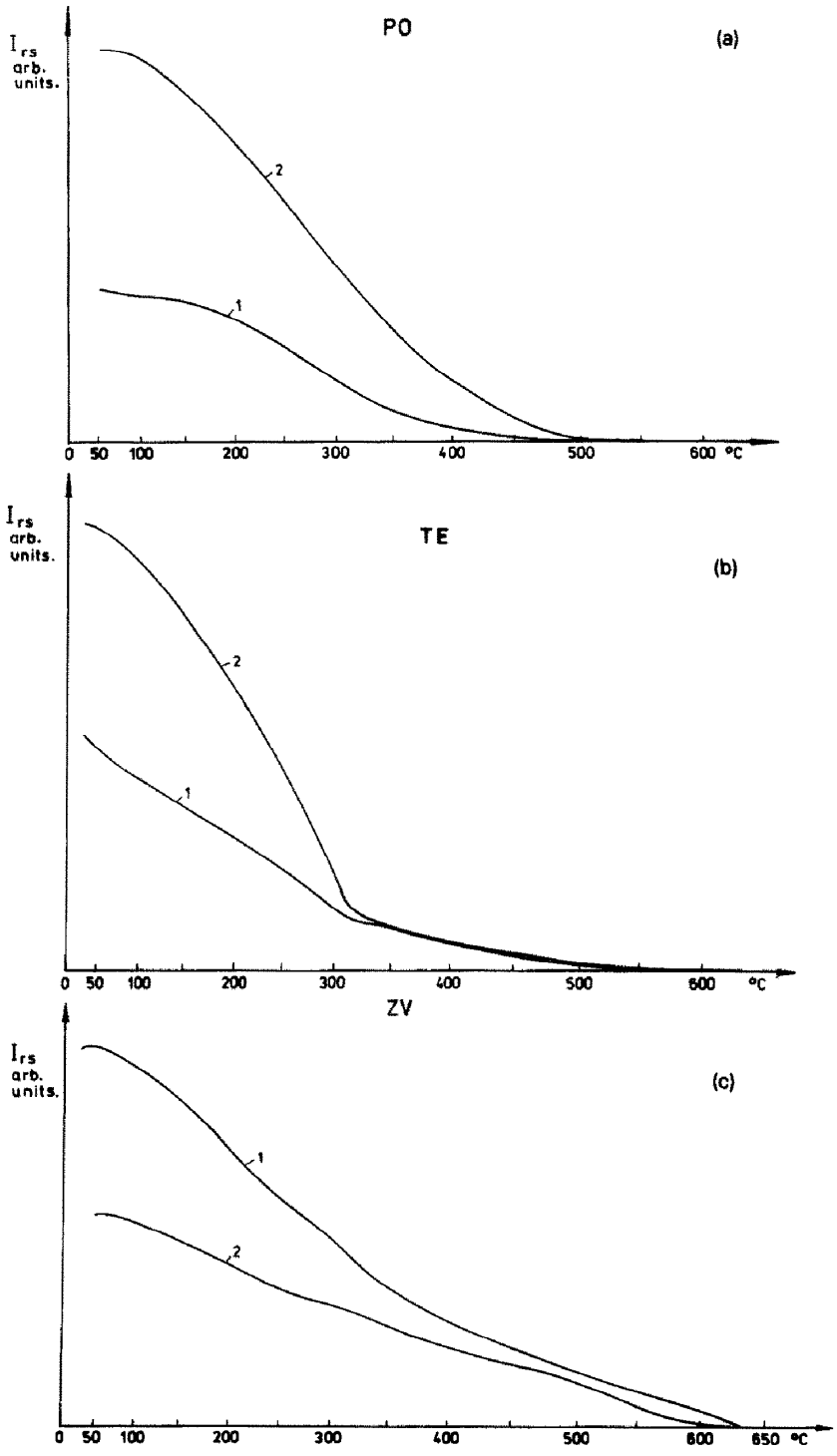


Fig. 2. Typical thermomagnetic curves. *Irs* = isothermal remanence; *T* = temperature, *1* = The first heating; *2* = the second heating. Letters above each plot denote names of exposures.

(see the legend to Fig. 1b). Results from the Lower Subtatric Nappe and Mala Fatra Mts. are taken into account in the last section of the paper.

Fifty oriented hand samples were collected at four exposures (13 sites) situated within the four nappes, and cylindrical specimens were cut out of them for standard thermomagnetic and paleomagnetic study. The rocks sampled in LT, BT and MS are from the Middle Jurassic, whereas the rocks from CH are of Lower Jurassic age.

Methods of measurement and analysis

The mineralogical analyses were carried out on several thin and polished sections per exposure. The sections were examined in order to identify the ore minerals present, using an ore microscope with a maximum resolution power of $1000\times$, which prevents identification of grains smaller than 1 μm. The thermomagnetic study led to the identification of magnetic minerals by their blocking temperatures T_b. The method consisted of thermal demagnetization of an isothermal remanence I_{rs} acquired in a field of 0.1 T, by heating up to 650°C in field-free space. The blocking temperatures of magnetic minerals present in the specimen appear as kinks on the thermomagnetic curves of I_{rs} vs T°C. The specimens were heated twice: first the I_{rs} was induced in the specimens in their natural state (I_{rs}–T) curve denoted by *1* in Fig. 2) and then I_{rs} was induced in the same specimens after the first heating–cooling cycle (I_{rs}–T) curve denoted by *2* in Fig. 2). The values of T_b related to curve *1* are characteristic for magnetic minerals present in the rock in its natural state, while those corresponding to *2* indicate magnetic minerals present in the rock after heating (see Kądziałko-Hofmokl and Kruczyk, 1987).

The paleomagnetic analyses were conducted on 303 specimens. In most cases they were demagnetized thermally in 50 or 100°C steps to 500–700°C in a non-magnetic furnace. Only a few of them were demagnetized using an alternating magnetic field. The remanence was measured with a JR-4 Czech spinner magnetometer (noise level 5×10^{-6} A/m). The characteristic remanence directions (ChRM) were determined from a least-squares linear fit to the demagnetization data (Kirschvink, 1980) and, at two sites, the remagnetization circles method of Halls (1978). The main criterion used in the ChRM determination was that of consistency of the directions from one specimen to another.

The magnetic susceptibility, K, and its anisotropy (AMS) were measured using a KLY-2 Czech bridge (noise level 10^{-8} SI) before thermal cleaning of specimens and after some heating steps. The values of the bulk susceptibility and AMS parameters were calculated using Jelinek's ANIZO 11 program (Jelinek, 1977).

Results

Mineralogical analysis

All thin sections revealed the presence of remnants of planktonic crynoidal (*Saccocoma*) facies. Radiolarian skeletons were found in radiolarites and radiolarian limestones. An unchanged framboidal synsedimentary pyrite was the only ore mineral identified with the ore microscope in the samples from the Choč Hills, the Low Tatra and the Belanski Tatra. In the Magura Spišska samples some of the framboidal pyrite grains were replaced by Fe-hydroxides.

Thermomagnetic analysis

The thermomagnetic analysis revealed the presence of magnetite, probably fine-grained according to its blocking temperature (about 450–500°C) and the shape of the first heating curve (Fig. 2a). In some samples, pyrrhotite with a blocking temperature of 320°C (Fig. 2b), or hematite with a blocking temperature of about 650°C (Fig. 2c) were present. The curves of the first heating for the MS samples do not show any evidence of Fe-hydroxides, but rather resemble the curves of fine-grained magnetite. The Fe-hydroxides identified by the microscopic analysis are either non-magnetic or poorly crystallized. According to Bagin et al. (1988), the blocking temperature of poorly crystallized goethite may attain 300°C. The curves of the second heating

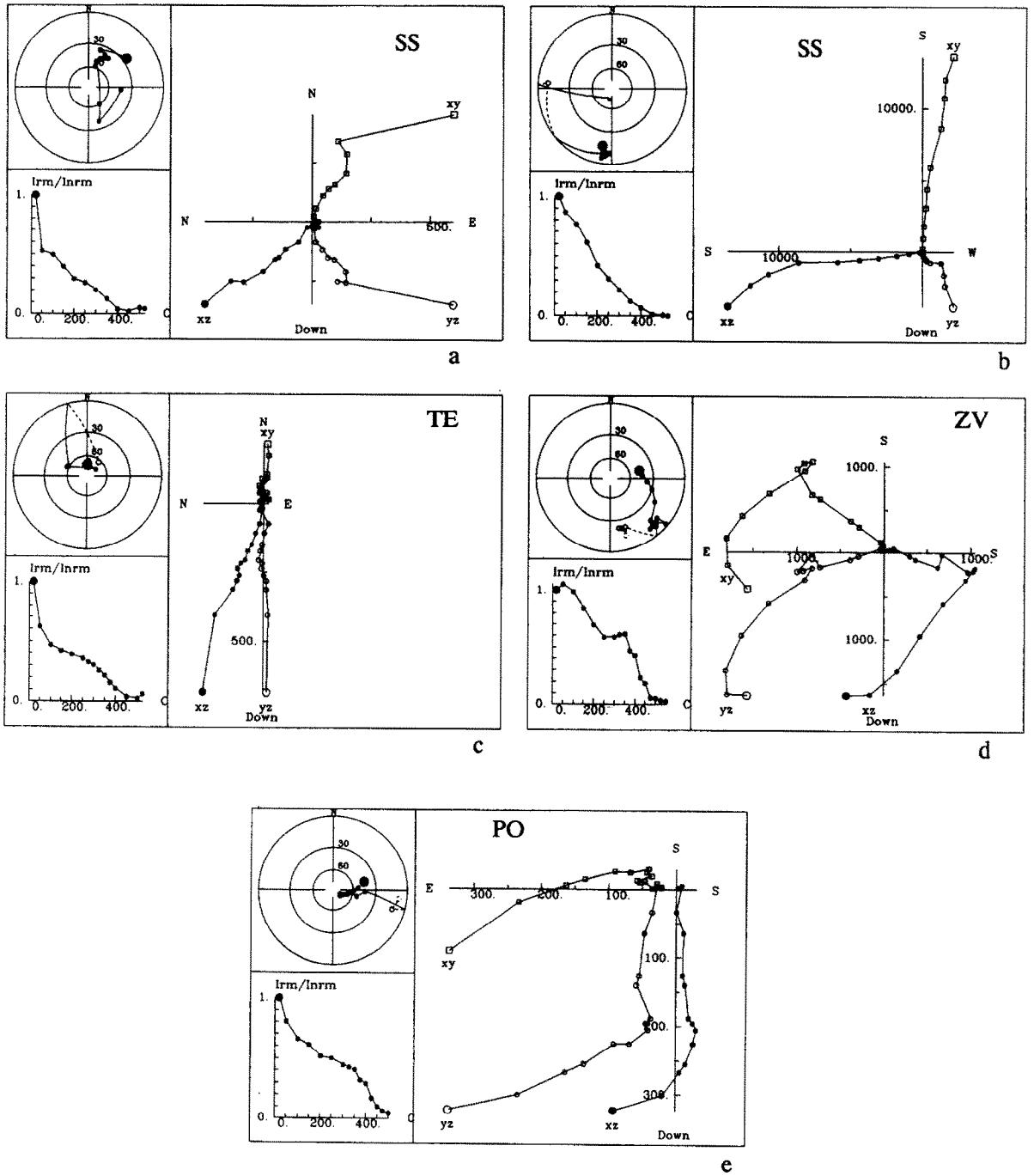


Fig. 3. Typical results of thermal demagnetization after bedding correction. Upper left-hand side: stereographic projections, full circles = lower hemisphere, open circles = upper hemisphere. Lower left-hand side: thermal decay of the NRM intensity; *Irm* = remanence after each heating step, *Inrm* = natural remanence before heating. Right-hand side: orthogonal plot of the three mutually perpendicular components. Large symbols denote the NRM intensity and direction before the beginning of the cleaning procedure. Letters above each plot denote names of the exposures.

TABLE 1

Jurassic paleomagnetic directions and pole positions for the Križna Nappe

Partial nappe	Exposure and bedding orientation	Age	n	N/N_0	bbc				abc							
					D°_m	I°_m	α°_{95}	k	D°_m	I°_m	α°_{95}	k	ΦºN	ΛºE	φ°_m	Polarity
Mala Fatra (*MF*) (Kądziałko-Hofmokl et al., 1990)	Zázriva (ZA) 40°, 320° Velka Branica (BV) 40°, 340°	Jm–u	21	173/193	328	85	4	66	321	44	3	96	58	253	26	N
Low Tatra (*LT*) $\varphi = 49.03$, $\lambda = 19.28$	Terlenska Dol. (TE) 20°, 355°	Jm	3	29/34	35	76	8	243	2	56	14	73	78	192	37	N
Belanski Tatra (*BT*) $\varphi = 49.25$, $\lambda = 20.20$	Ždiarska Vidla (ZV) 45°, 25°	Jm–u	4	17/33	190	73	12	58	40	59	8	118	61	113	40	n
Magura Spišska (*MS*) $\varphi = 49.28$, $\lambda = 20.53$	Podoliniec (PO) 55°, 90°	Jm	3	12/7	312	70	24	28	75	46	12	114	30	102	28	N
Choč Hills (*CH*) $\varphi = 49.12$, $\lambda = 19.23$	Skladana Skala (SS) 25°, 15°	Jl	3	15/26	116	80	12	104	39	63	12	104	63	105	44	N
Western Tatra (*LSN*) (Kądziałko-Hofmokl and Kruczyk, 1987)	40°, 20°	Jm–u	7						22	59	4.3	198	71.7	132.2		
Cracow Upland (*CU*) (Kądziałko-Hofmokl and Kruczyk, 1987)		Jm–u	(8)						17	56	7.3	58	72.3	150.4		M
Eurasia (*EA*) (Westphal et al., 1986)		Jm–u	(4)						25	59	7	187	70	128		
France (1, Normandy)		Jl(Plien.)							29	61			68.5	117.9		
(2) (1 and 2 Galbrun and Daly, 1987)		Jl(Toar.)							30	65	5	109	70.1	102.6		M

φ, λ = Geographic coordinates of the partial nappes; n = number of sites; N = number of specimens taken in the statistics; N_0 = number of specimens demagnetized; D_m, I_m = mean declination, mean inclination before (bbc), after (abc) bedding correction; α_{95}, k = Fisher statistical parameters; ΦN, ΛE = latitude and longitude of the paleomagnetic; φ_m = paleolatitude.

indicate an increasing amount of magnetite (Fig. 2a,c) and pyrrhotite (Fig. 2b), or the appearance of pyrrhotite due to heating. The pyrrhotite observed in the fresh material (Fig. 2b) is probably of post-pyrite origin.

Susceptibility

The bulk susceptibilities K measured before heating of the specimens were rather low, and remained in the range of $50–120 \times 10^{-6}$ SI at most sites. In the Kimmeridgian rocks from Belanski Tatra, the susceptibility attained values of $2–70 \times 10^{-6}$ SI, and in Magura Spišska it was negative (due to diamagnetic minerals) and ranged from -2 to -30×10^{-6} SI. During heating, K behaved similarly in all specimens, its value decreasing by 10–20% between 300 and 500°C, and then increasing considerably due to new magnetic minerals appearing during heating.

The magnitude of the AMS observed in rocks from the Low Tatra, the Choč Hills, and the Belanski Tatra is low, as the anisotropy parameter, $P = K_{max}/K_{min}$, did not exceed 1.03 in most cases. The magnetic fabric, although not well developed, has a depositionary character, as the axis of minimum susceptibility is perpendicular to the sedimentation plane. Heating up to 500–600°C resulted in a slightly increasing magnitude of AMS, which remains characteristic of sedimentary rocks (Jeleńska and Kądziałko-Hofmokl, 1990). In the rocks from Magura Spišska, the AMS is very weak.

Paleomagnetic study

All paleomagnetic results obtained for the investigated nappe fragments are listed in Table 1, as are the corresponding data for Mala Fatra and the Lower Subtatric Nappe. The table also contains some reference data for the Eurasian Middle–Upper and Lower Jurassic. The ChRM directions for individual sites and exposures were isolated using the criterion of the highest repeatability. The most typical demagnetization results for each nappe fragment are shown in Figure 3.

In the Choč Hills two types of NRM behavior were observed. The first type (Fig. 3a) corresponds to a two-component remanence. The component with a higher T_b and well grouped directions within the exposure is probably carried by magnetite. The directions of the softer components are scattered. Other demagnetization results reveal only one component of remanence with highly scattered directions (Fig. 3b). It is perhaps carried by the secondary pyrrhotite visible on thermomagnetic curves.

In the specimens from the Low Tatra, the isolated ChRM is represented either by the line fitted to the whole demagnetization curve or to its part ranging from 20 to 300°C (Fig. 3c). Its carriers are either magnetite or secondary pyrrhotite, both magnetized in the short time span in the field of the same, or similar, directions.

The remanence of the Belanski Tatra specimens is the resultant of the two components. One of them, isolated as the ChRM, has a T_b of 300–350°C and is probably carried by magnetite. Its directions group tightly within the Bathonian and the Kimmeridgian. The directions of the second, harder component are highly scattered, especially in the Bathonian rocks. The NRM directions during the demagnetization procedure lie on great circles (Fig. 3d). Halls's (1978) method of remagnetization circles failed to find the direction of the remagnetization field. The Oxfordian radiolarites from Belanski Tatra yielded no reliable results.

The specimens from Magura Spišska (Fig. 3e) were heated to 600–700°C. In most cases the demagnetization results show the presence of two components with distinctly different T_b spectra. In two-thirds of the specimens, the component treated as ChRM appears either in the temperature range of 300–500°C or 50–500°C. In the remaining specimens, a component of the same direction appeared in the range of 20–325°C. It is supposed that the ChRM of those rocks is carried by magnetite and/or Fe-hydroxides, magnetized in the field of the same direction and therefore within a short time interval. The directions of the second component are very strongly scattered, and are probably carried by some post-Fe-hydroxide minerals.

The isolated ChRM is of normal polarity in all

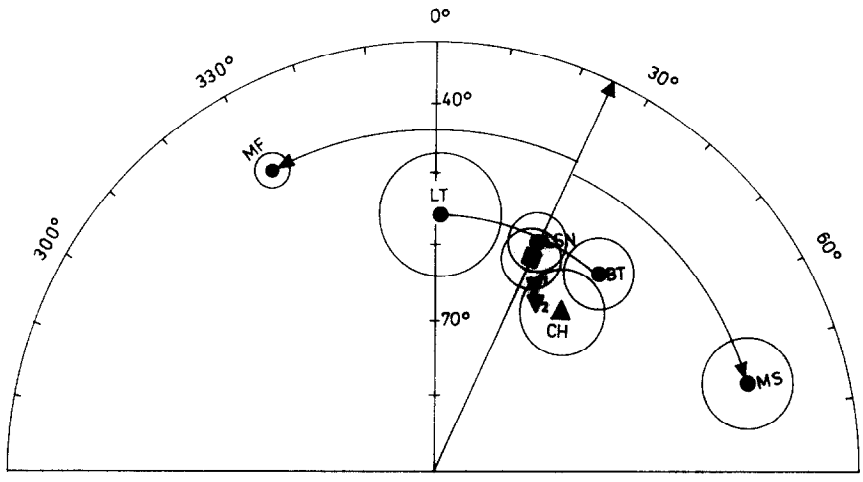


Fig. 4. Mean directions of the ChRM after tectonic correction for six nappe fragments, together with their confidence circles in comparison with the reference Middle–Upper Jurassic paleodeclination denoted by an arrow, and reference directions. The partial nappes are denoted as in Fig. 1 and Table 1: square = Eurasian reference direction (combined from the Jm and Ju data of Westphal et al., 1986). *1* = Pliensbachian, Normandy; *2* = Toarcian, France (*1* and *2* after Galbrun and Daly, 1988). Declination deviations for individual nappes are shown by arcuate arrows.

cases. The mean directions before bedding correction (bbc) and after bedding correction (abc) are included in Table 1, together with the parameters of the Fisher statistics and pole positions computed from the abc directions. The table also comprises the reference directions and pole positions for the Cracow Upland, stable Poland (Kądziałko-Hofmokl and Kruczyk, 1987), the Eurasian Middle–Upper Jurassic pole position and the direction computed from it for the region considered (Westphal et al., 1986), and two results of the Lower Jurassic from France (Galbrun and Daly, 1987). The bbc directions obtained in this study do not resemble either the Jurassic or the post-Jurassic Eurasian reference directions (see Westphal et al., 1986). We surmise, therefore, that the ChRMs isolated for the Križna Nappe fragments are pre-tectonic.

The mean ChRM directions obtained after bedding corrections are shown in Fig. 4. The figure includes the reference data for Eurasia for the Middle–Upper Jurassic and two Lower Jurassic data from France. The reference declination (EA in Table 1) is marked by an arrow. The directions obtained for the Križna Nappe fragments lie along a band spanning inclinations from 44 to 63° and declinations from 320 to 75°. They form a fan-like pattern similar to that formed by vergencies (see Fig. 1b).

In order to determine the origin of the arcuate shape of the region studied, the declination deviations $(D_r - D)$ and the respective vergency deviations $(V_r - V)$ were compared for each site: D_r is the reference declination (taken from Westphal et al., 1986), D is the declination obtained for each investigated site, V_c is the general northward trend of the thrust (the reference vergency; see, e.g., Bely and Fusan, 1967) and V is the vergency of each fold observed in the field.

Discussion and conclusions

The results (Table 1 and Fig. 4) shows that the inclinations of ChRM (after bedding correction) found for the nappe fragments, comprising the Middle–Upper Jurassic rock, lie in the range of 44–59°. The inclinations obtained for the two extreme nappes, Mala Fatra (MF) in the west and Magura Spišska (MS) in the east, differ considerably from the reference inclination, and no explanation is available for this. The inclinations for the Low Tatra (LT), the Belanski Tatra (BT) and the Lower Subtatric Nappe (LSN) are in good agreement with the reference data. The inclination obtained for the Lower Jurassic Choč Hills (CH) rocks is close to the results of Galbrun and Daly (Table 1). The declinations of the ChRM

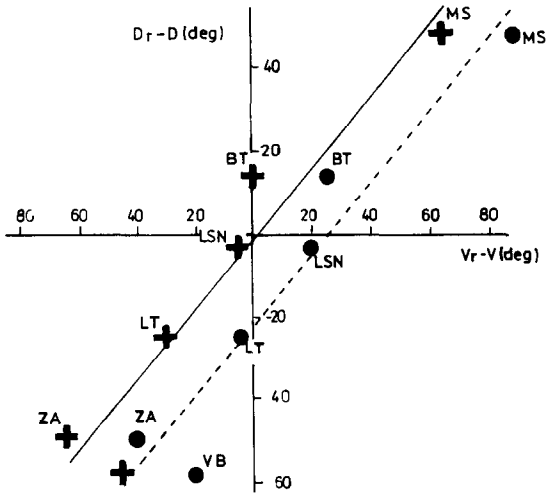


Fig. 5. Plot of declination deviation $D_r - D$ and the vergency deviation $V_r - V$ for the Middle–Upper Jurassic exposures. Solid symbols and the dashed line correspond to the northward trend of overthrusting. Crosses and the continuous line correspond to the trend of overthrusting deviated by 20° to the east. Letters denote the exposures (see Table 1).

obtained for the Middle–Upper Jurassic differ from each other and from the reference declination $D_r = 25°$ (declination for Eurasia (EA) in Table 1). The only exception is LSN, for which the declination is practically the same as D_r. The most deviated are the declinations of the westernmost MF and easternmost MS nappe fragments. The deviations of the LT and BT declinations from D_r, although smaller, also exceed the confidence circles. The declination of the CH rocks is close to the French Lower Jurassic results. The results presented indicate that the investigated regions rotated against the stable Eurasian Plate in the following way: MF, 60° anticlockwise, LT, 20° anticlockwise; BT, 20° clockwise; MS, 50° clockwise. The LSN and CH nappe fragments seem to be unrotated.

The observed agreement between the declinations and vergencies allows us to answer the question as to the origin of the curvature of the Inner Carpathian Arc. In Fig. 5 in presented the plot of the declination deviations $D_r - D$ against the vergency deviations $V_r - V$, where $D_r = 25°$ and $V_r = 0$. Solid circles and a dashed line correspond to the northward trend, which crosses and a continuous line correspond to the trend of overthrusting of about 20°E. The straight lines fitted to the data have a slope coefficient of about 0.9 (the number of data was too small to calculate the correlation coefficient or the accuracy of fitting). This result strongly suggests oroclinal bending of the Inner West Carpathians, which originally formed a straight belt. This conclusion is in agreement with the suggestions of Bazhenov and Burtman (1990). According to our results, the bending of the region studied took place after the overthrusting of the Križna Nappe, and the trend of the overthrust was not purely northward but slightly deviated to the east.

Acknowledgments

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